— Deep Learning —

Remote Access to the Lab Computers

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Instructions for Remote Access to the Lab Computers

If you would like to work on the assignments remotely, and can't or don't want to set up your own computers for deep learning, you can use the CS lab computers remotely.

- Select a computer that is not in use (since you will likely make heavy use of its GPU)
  - Any of the Linux-based computers listed here that have Geforce GPUs should work well:

    https://support.cs.umu.se/labs/
  - Check which computes are free here:

    https://support.cs.umu.se/system/status
  - Or, after logging in, type

    ```
    $ users
    ```

    to see if there's anyone else currently logged in to that machine
  - Avoid using computers that are in use by someone else!
- Open a terminal on your computer, and type:

```
$ ssh <username>@<hostname>.cs.umu.se
```

- Type your password and press enter (you will likely not see any characters while typing the password)
- You are now logged in remotely, and can run your Python code after activating the conda environment. See instructions for that here.

Remote Access to Jupyter Notebooks via SSH

On your local computer, you can edit and run Python scripts that are actually residing on the CS lab computers. We will detail this for Jupyter notebooks, but it is also possible to do this directly in PyCharm and Visual Studio Code among many other.

So, you can open and manipulate a Jupyter Notebook that is actually running on a remote computer. This can be achieved by opening an SSH tunnel to the remote computer first. The communication between your local browser and the remote Jupyter Notebook will be passed through the tunnel (with strong encryption), and you can work just as if you were sitting in front of that remote computer in the lab.

If you have a Linux-based or MacOS Computer

- Open a terminal
- Connect and open the tunnel by typing:

  ```
  ssh -L 8080:localhost:8080 <username>@<hostname>.cs.umu.se
  ```

- Type your password and press enter
- Activate the Conda environment:

  ```
  $ conda-activate
  ```

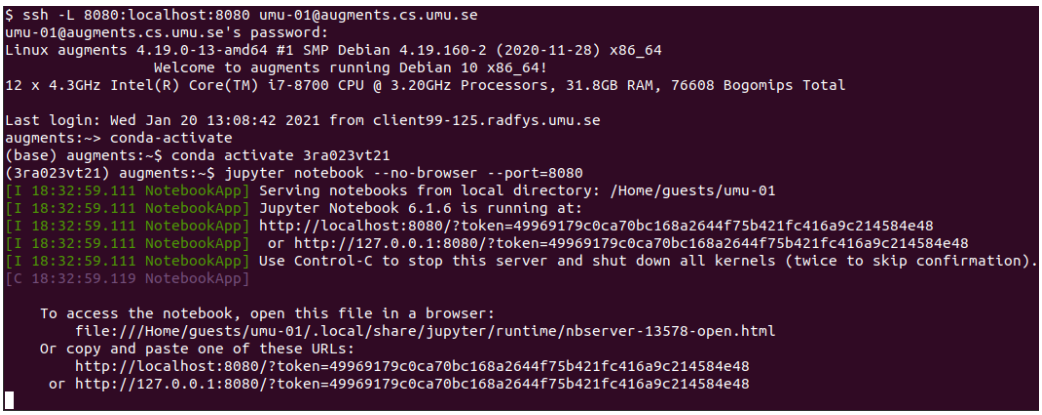
  ```
  $ conda activate 5dv236vt26
  ```

- Navigate to the directory containing your Jupyter Notebook:

  ```
  $ cd <path>
  ```

- Start Jupyter Notebook on port 8080, without starting a browser:

  ```
  $ jupyter notebook --no-browser --port=8080
  ```



- The terminal will print some text. Part of it is a *token* that we need later. Copy the token. The token in the example image above was

  `49969179c0ca70bc168a2644f75b421fc416a9c214584e48`

- Open an internet browser (*e.g.*, Firefox) and navigate to:

  `http://127.0.0.1:8080/`

- You will be prompted to enter the token that you copied earlier. Paste the token and log in

- Done! You are now running a Jupyter Notebook on the remote computer, but controlling it from your local computer

If you have a Windows-Based Computer

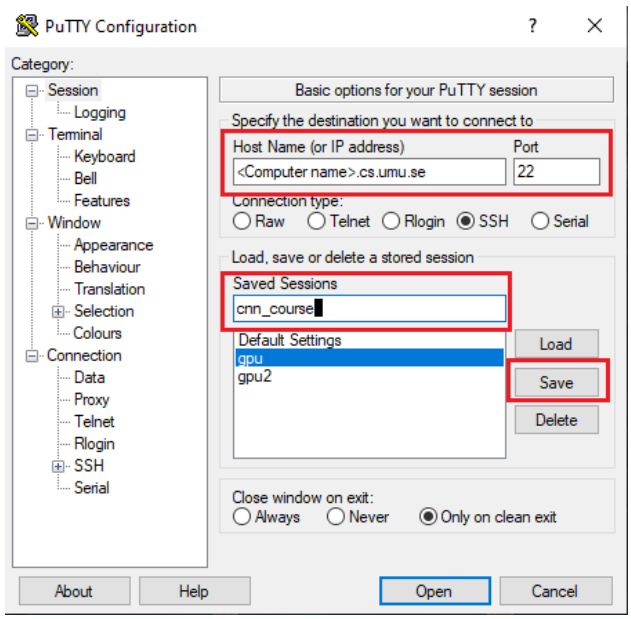
- If your Windows has built-in SSH client, open a Command Prompt and follow the instructions for Linux-based and MacOS computers above

- Unless you already have an SSH client, you can download PuTTY:

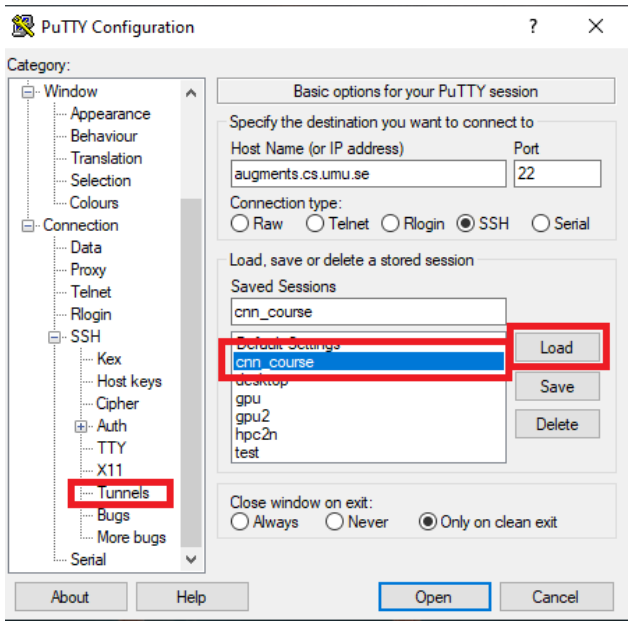
  https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html

- Start PuTTY to configure the SSH Tunnel

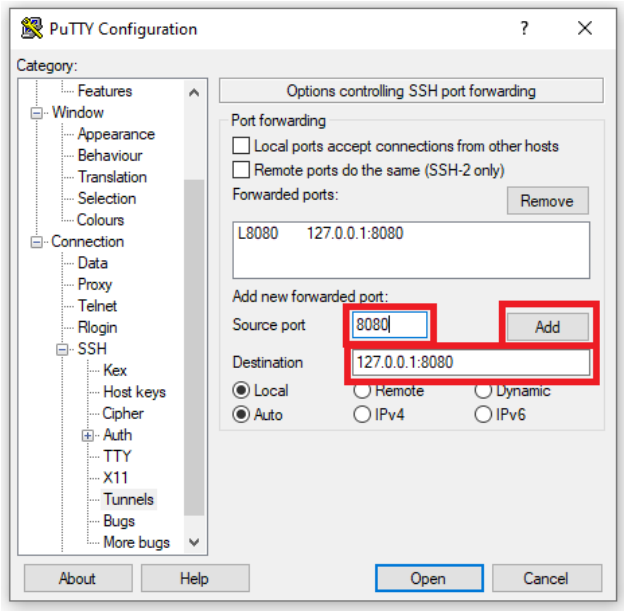
- Fill in the fields in the red rectangles and save it as a session (for later reuse):



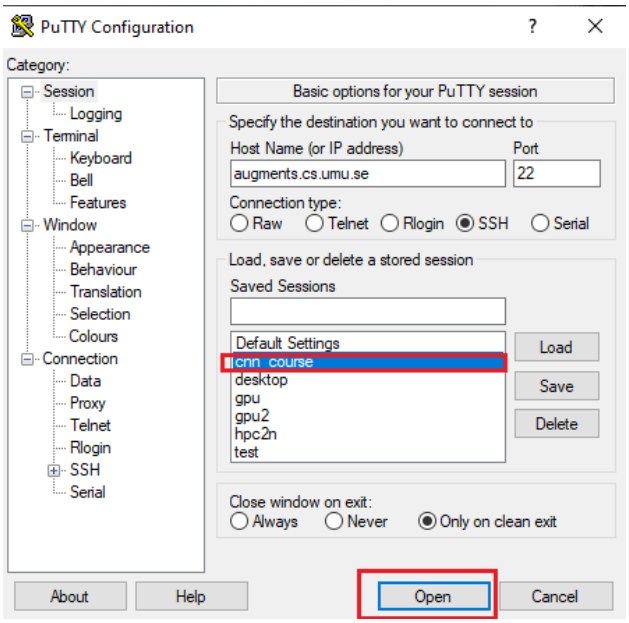
- Load the session, click on *Tunnels* under *SSH* in the left *Category* pane:



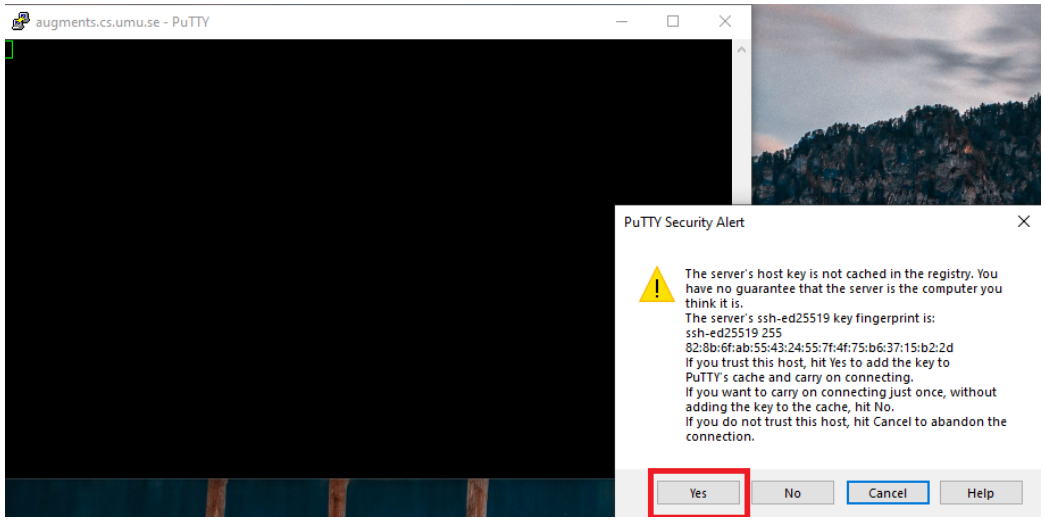
- Fill in the *Source port* and *Destination* fields, and click *Add*:



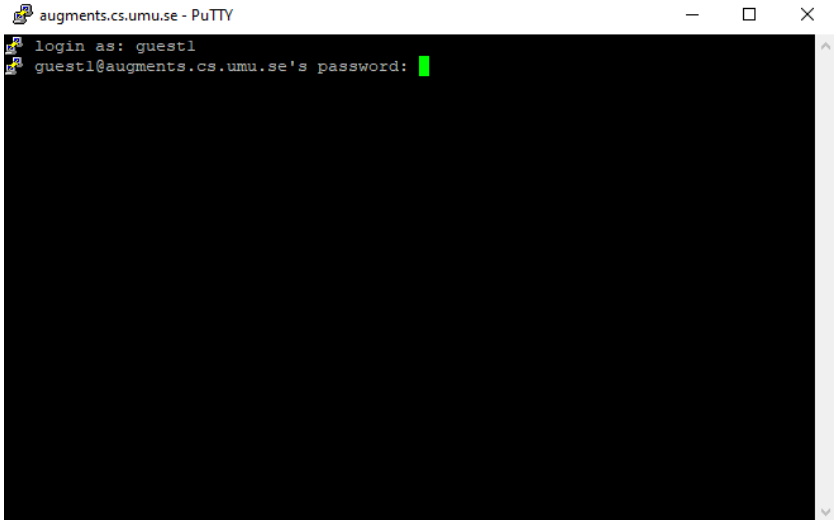
- Go back to *Session* in the left *Category* pane and save the session
- To open a PuTTY terminal, double-click on the saved session:



- You will be prompted to accept the encryption fingerprint
- If this is the correct fingerprint, click *Yes*:



- Enter your CS username and password:



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Don't hesitate to contact us if you have any questions!

Good luck, and have fun!